**Quantitative Finance > Computational Finance**

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# An Advanced Ensemble Deep Learning Framework for Stock Price Prediction Using VAE, Transformer, and LSTM Model

Anindya Sarkar, G. Vadivu

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This research proposes a cutting-edge ensemble deep learning framework for stock price prediction by combining three advanced neural network architectures: The particular areas of interest for the research include but are not limited to: Variational Autoencoder (VAE), Transformer, and Long Short-Term Memory (LSTM) networks. The presented framework is aimed to substantially utilize the advantages of each model which would allow for achieving the identification of both linear and non-linear relations in stock price movements. To improve the accuracy of its predictions it uses rich set of technical indicators and it scales its predictors based on the current market situation. By trying out the framework on several stock data sets, and benchmarking the results against single models and conventional forecasting,

the ensemble method exhibits consistently high accuracy and reliability. The VAE is able to learn linear representation on high-dimensional data while the Transformer outstandingly perform in recognizing long-term patterns on the stock price data. LSTM, based on its characteristics of being a model that can deal with sequences, brings additional improvements to the given framework, especially regarding temporal dynamics and fluctuations. Combined, these components provide exceptional directional performance and a very small disparity in the predicted results. The present solution has given a probable concept that can handle the inherent problem of stock price prediction with high reliability and scalability. Compared to the performance of individual proposals based on the neural network, as well as classical methods, the proposed ensemble framework demonstrates the advantages of combining different architectures. It has a very important application in algorithmic trading, risk analysis, and control and decision-making for finance professions and scholars.

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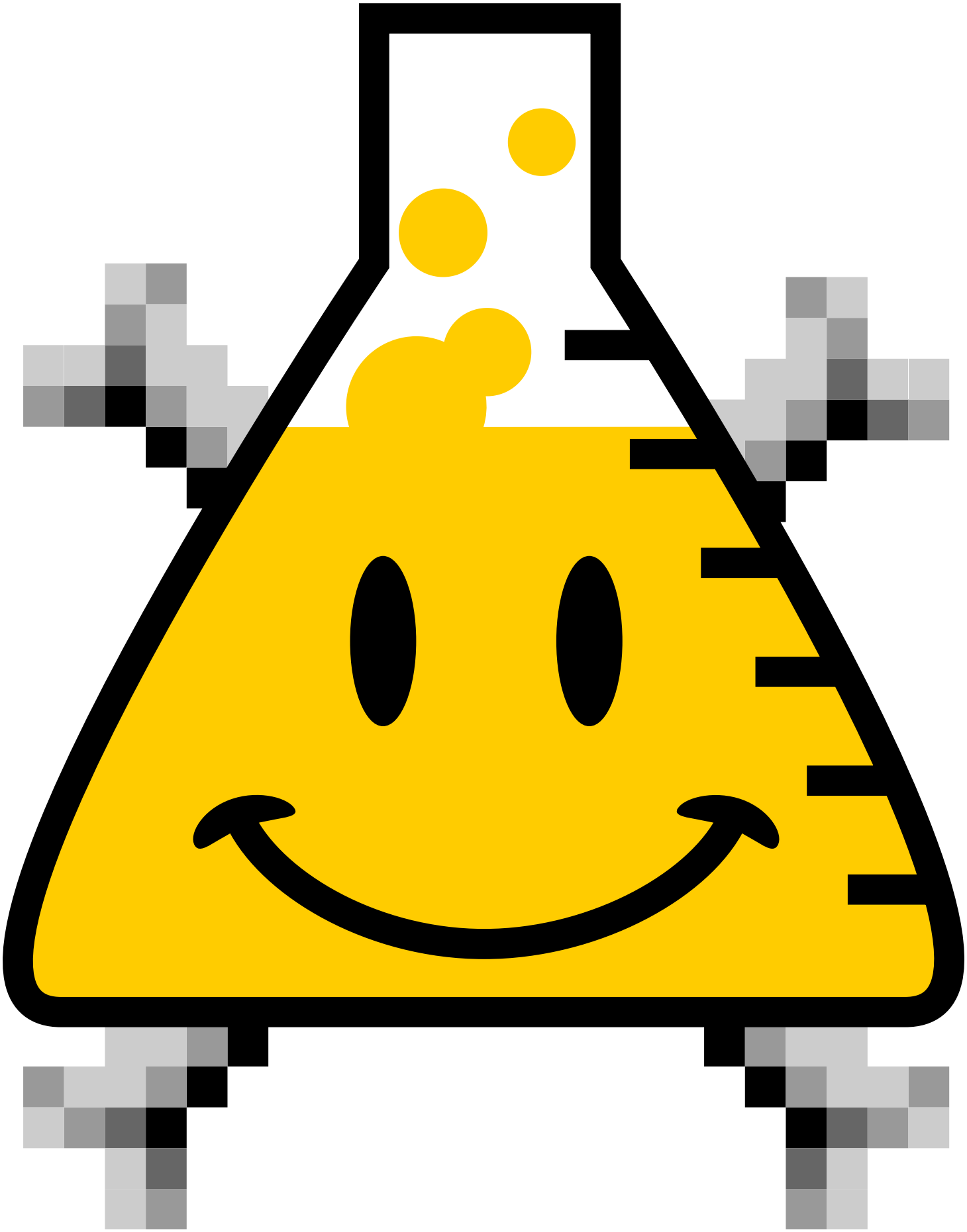

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